Numerical model and data companion for acoustic wave chaos in a periodic phononic-crystal waveguide

This companion connects the numerical model to the data used in the sixteen result figures of the article and supplement. Five method diagrams retain the physical and computational context, while the text concentrates on provenance, field reconstruction, array coordinates, uncertainty records and figure-to-file correspondence. Complementary relations specify the analytic benchmark and the interpretation of stored field arrays. Each result is linked to its figure directory, inputs and plotting operations. Terminal instructions describe the selection, display and PDF export of the saved numerical results.

I. ROLE OF THE MODEL COMPANION

The `PnCW` package contains processed numerical data and Python scripts for main-article Figs. 3–10 and Supplemental Material (SM) Figs. S1–S8. The package is publicly available at https://github.com/hiperezag/PnCW. This document explains how the model, numerical methods and saved records connect to those figures, complementing the article and supplement. The five diagrams provide the physical and computational context for the numerical-result directories. Main-article Figs. 1 and 2 show the waveguide geometry and the geometric construction of the ray-incidence bound; these schematics remain in the article, whereas this companion documents the sixteen numerical-result figures and their associated data and plotting folders.

Diagram D1 separates three representations of the same guide. Geometric ray calculations supply the dynamical certificate and collision sample. The boundary integral-equation method (IEM) provides representative pressure modes and wall traces. Spectral elements (SEM) provide the validated finite-window ensembles used for spectral, spatial and energy analyses. The plotting package reads these outputs, assembles the panels and exports the numerical-result figures. In Diagram D1, h_{top} is topological entropy per inclusion collision. The article supplies the physical argument and principal observables. SM Secs. S.I–S.III document IEM, rays and SEM validation; Secs. S.IV–S.VI give complete spectral tests,

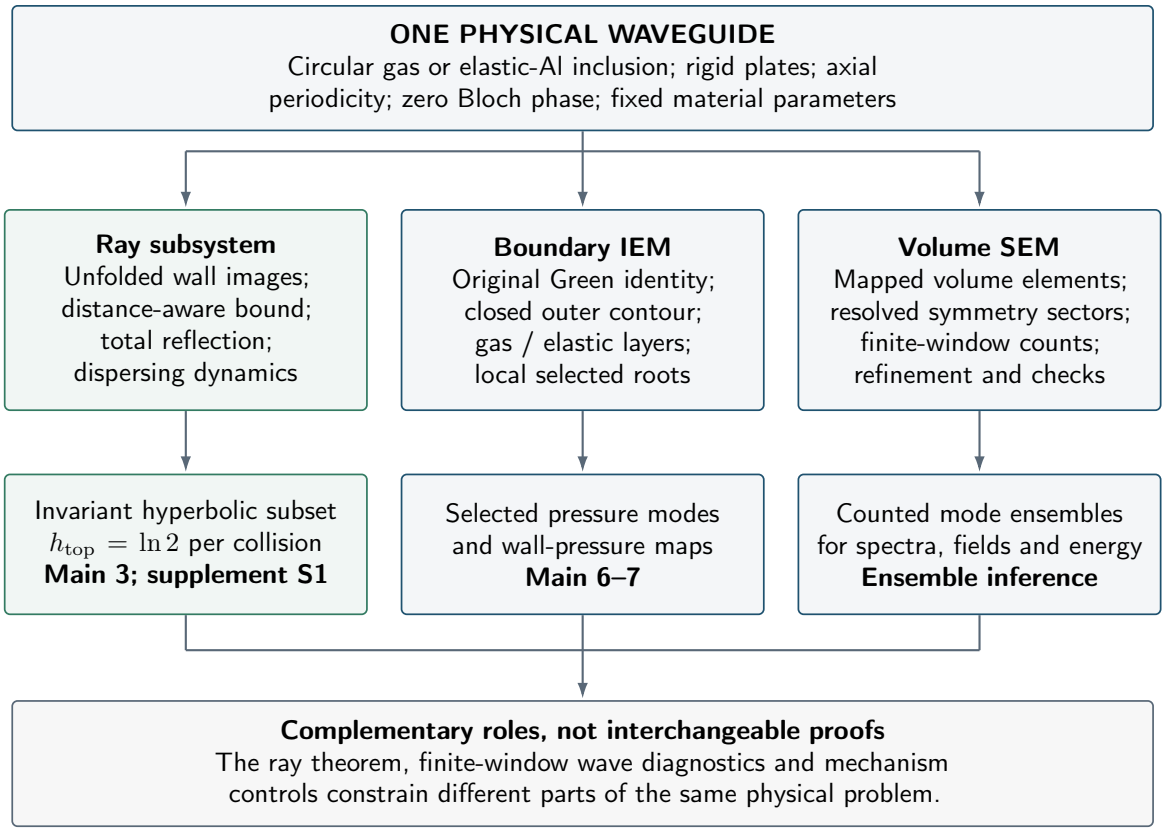


Diagram D1. One physical problem with three complementary representations. The ray certificate is geometric; IEM reconstructs selected modes and wall traces; SEM supplies counted inference ensembles. Analytic integrable controls and the independent local planar-interface calculation retain their separate roles.

spatial checks and mechanism controls. The additional equations here specify the analytic benchmark, sampling coordinates and numerical data layout.

II. PHYSICAL SETTING AND NOTATION FOR THE DIAGRAMS

The model is a two-dimensional, zero-Bloch-phase cell of an axially periodic guide between rigid plates. Its circular inclusion contains fast gas or isotropic elastic aluminum, with plane strain in the solid. Wave ensembles and representative fields use inclusion area fraction $f = 0.03$; the ray certificate also covers $f = 0.01, 0.06, 0.10$. The transverse direction is bounded by plates, not physically Bloch-periodic. The square arrangement of unfolded ray images represents this same guide. Diagram D2 distinguishes walls, paired axial cuts and material coupling. Detailed equations and material values remain in article Sec. II.

The diagrams use axial period P, plate separation b ($P = b = L_{\mathrm{phys}}$) and axial Bloch wavenumber $K = 0$. Pressure p, displacement $\mathbf{u}$ and Cauchy stress $\boldsymbol{\sigma}$ use the article's air-based reduction: pressure and stress are scaled by $\rho_a c_a^2$ and displacement by the length unit ℓ_0. The labels a, g, s denote air, gas and solid; ρ_a, ρ_g are the fluid densities and c_a is air sound speed. The unit normal $\hat{\mathbf{n}}_s$ points from the inclusion into air; ∂_{n_s} is differentiation along it and ∂_y differentiation transversely. Ω is reduced frequency.

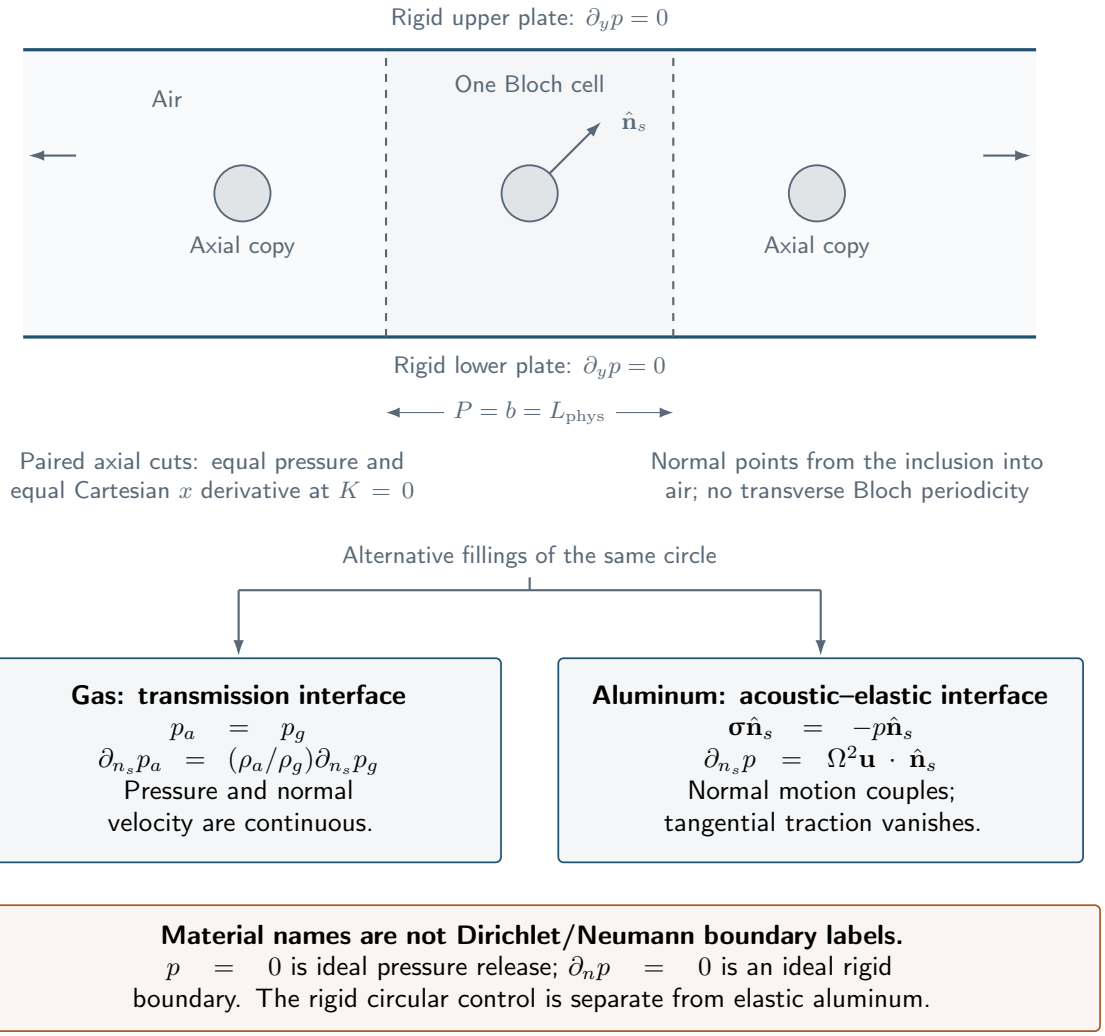


Diagram D2. Boundary-value specification, shown schematically in two dimensions. The three circles belong to one collinear periodic row. Dashed cuts delimit one axial cell, while the upper/lower lines are rigid plates, not transverse periodic edges. The lower boxes give alternative gas and aluminum interface conditions; the normal points into air. Pressure, displacement and stress use the reductions defined in Sec. II; the gas density ratio is unchanged by the common pressure/length scaling.

The empty square has two roles: its analytic modes are an integrable physical control, and agreement with an inclusion-free IEM solve benchmarks the homogeneous-cell method. A rigid circular inclusion is a different mechanism control. It is neither the empty square nor a change of aluminum's material label into a Neumann condition. Scalars are italic, vectors upright bold, matrices sans-serif and unit vectors hatted; n in an ideal-boundary label denotes its normal.

III. RAY AND SPECTRAL RECORDS

A. Geometric evidence

Figure 3 is generated by `F_03_ray_certificate`. Its script plots the certified incidence bounds and material critical angles, the stored median and minimum–maximum periodic-orbit instability ranges, and the symbolic adjacency matrix. These panels connect the total-reflection margin, sampled instability and symbolic dynamics described in article Sec. III and SM Sec. S.II.A.

Figure S1 is generated by `S_01_periodic_ray_section`. The script assembles the unfolded geometric view from saved circle centers and itinerary vertices, draws the directed triangle, and displays the collision coordinates with the material and certified thresholds. The finite sample contains 336 records from 58 itineraries at filling fraction $f = 0.03$, including coincident points, repetitions and reversed orientations (SM Sec. S.II.B). The ray coordinates describe inclusion collisions; the wall maps in Fig. 7 describe pressure-trace position and direction.

B. Numerically accepted ensembles

Diagram D3 retains the distinction between counting, extracting and validating a catalogue. The independent method of particular solutions (MPS) checks predetermined samples where documented; sample coverage is not extended to every mode. Numerical acceptance occurs upstream of this package and does not depend on an irregular-looking field or agreement with a random-matrix reference. In Diagram D3, λ_j is a squared-frequency eigenvalue, Ω_j

its positive reduced-frequency root and j the level index within a symmetry sector. MAC denotes the modal assurance criterion used in refinement comparisons.

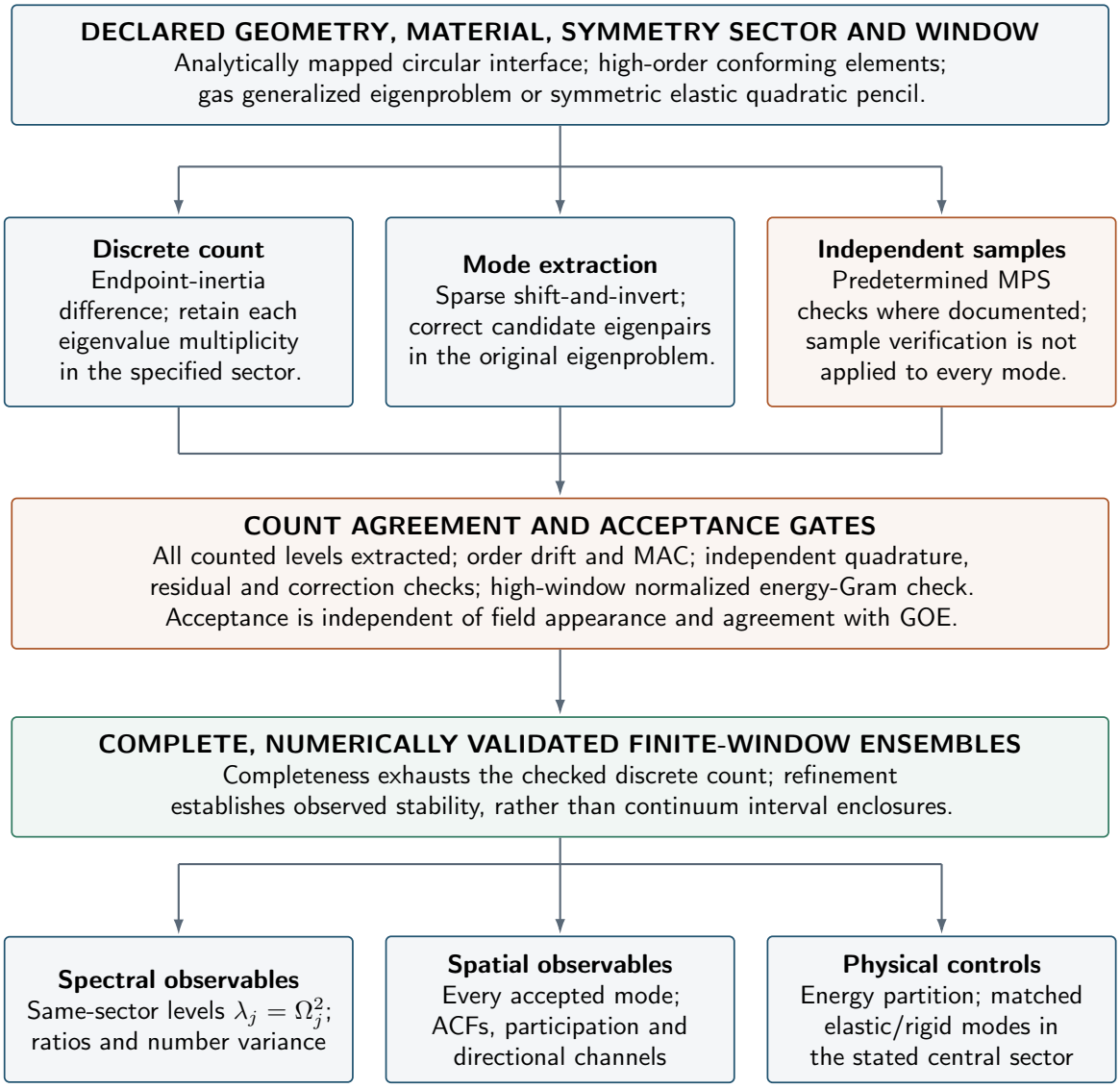


Diagram D3. From a declared SEM eigenproblem to accepted finite-window catalogues. Count agreement concerns the discrete problem with multiplicity; refinement and independent checks establish observed numerical stability. MPS checks cover predetermined samples where documented, not every mode; the separate rigid control has no independent MPS sample. The accepted ensembles, not representative visual examples, supply spectral and complete-ensemble spatial inference.

Figure S2 is generated by `S_02_numerical_validation`. The script groups the stored frequency-order drifts and MPS comparisons by material and plots the pressure and connected squared-pressure grid changes against their acceptance limits. It uses six order-check records,

ten fixed MPS comparisons and 144 modal grid checks per observable in $60 < \Omega \leq 61$, ppa (SM Sec. S.III.C).

Figure 4 is generated by `F_04_spectral_scale_contrast`. The script selects the five central aluminum sectors from `shared/central_spectrum.json`, then plots the mean adjacent-gap ratio and the affine number variance at $L_s = 8$ against matched finite-reference quantiles. Here L_s is interval length in unfolded mean spacings. The common window is $26 < \Omega \leq 34$ (article Sec. V and SM Sec. S.IV.B).

Figure S3 is generated by `S_03_number_variance`. It uses the same shared record to display all five sectors at $L_s = 1, 2, 4, 8$, with separate affine and monotone-quadratic rows, physical number-variance curves, reference medians and pointwise 2.5–97.5% reference bands. The shared file's `rows` contain physical statistics and calibrated decisions, and `reference_quantiles` contains finite-reference display bands. The reference lookup is indexed by level count, reference model and statistic. GOE denotes the Gaussian orthogonal ensemble. Formal decisions follow the calibrated tests in SM Sec. S.IV; the plotted bands describe pointwise finite-reference variability.

Figure 5 is generated by `F_05_high_frequency_spectra`. The script selects cumulative-distribution-function (CDF) and number-variance coordinates from `curves.csv`, draws the exact physical empirical CDF steps and reference curves, and reads the material level counts, mean ratios and Holm-adjusted probabilities from `data.json`. Its $58 < \Omega \leq 62$ ppa catalogues contain 185 gas and 184 aluminum levels. The lower panels display both unfoldings at the four tested interval lengths (article Sec. V and SM Sec. S.IV.D). Expanded and nested windows retain their respective sample sizes and calibrated decisions; their overlap provides the sensitivity comparison in SM Sec. S.IV.E.

IV. REPRESENTATIVE IEM MODES AND RECONSTRUCTION

A. Boundary calculation and common identities

Diagram D4 preserves the closed-contour Green-identity basis, local singular-value-decomposition (SVD) root search and separate physical checks, as developed in article Sec. II and SM Sec. S.I. The resulting pressure atlas is generated by `F_06_pressure_modes`; its matching wall maps are generated by `F_07_wall_husimi`. Both scripts use the same

panel identities in `shared/representative_modes.json`. In Diagram D4, $\mathsf{A}_{\mathrm{IEM}}(\Omega)$ is the frequency-dependent homogeneous boundary matrix, and $\sigma_{\min}, \sigma_{\max}$ are the smallest and largest singular values after column equilibration. These scalar singular values are distinct from the Cauchy-stress tensor $\boldsymbol{\sigma}$.

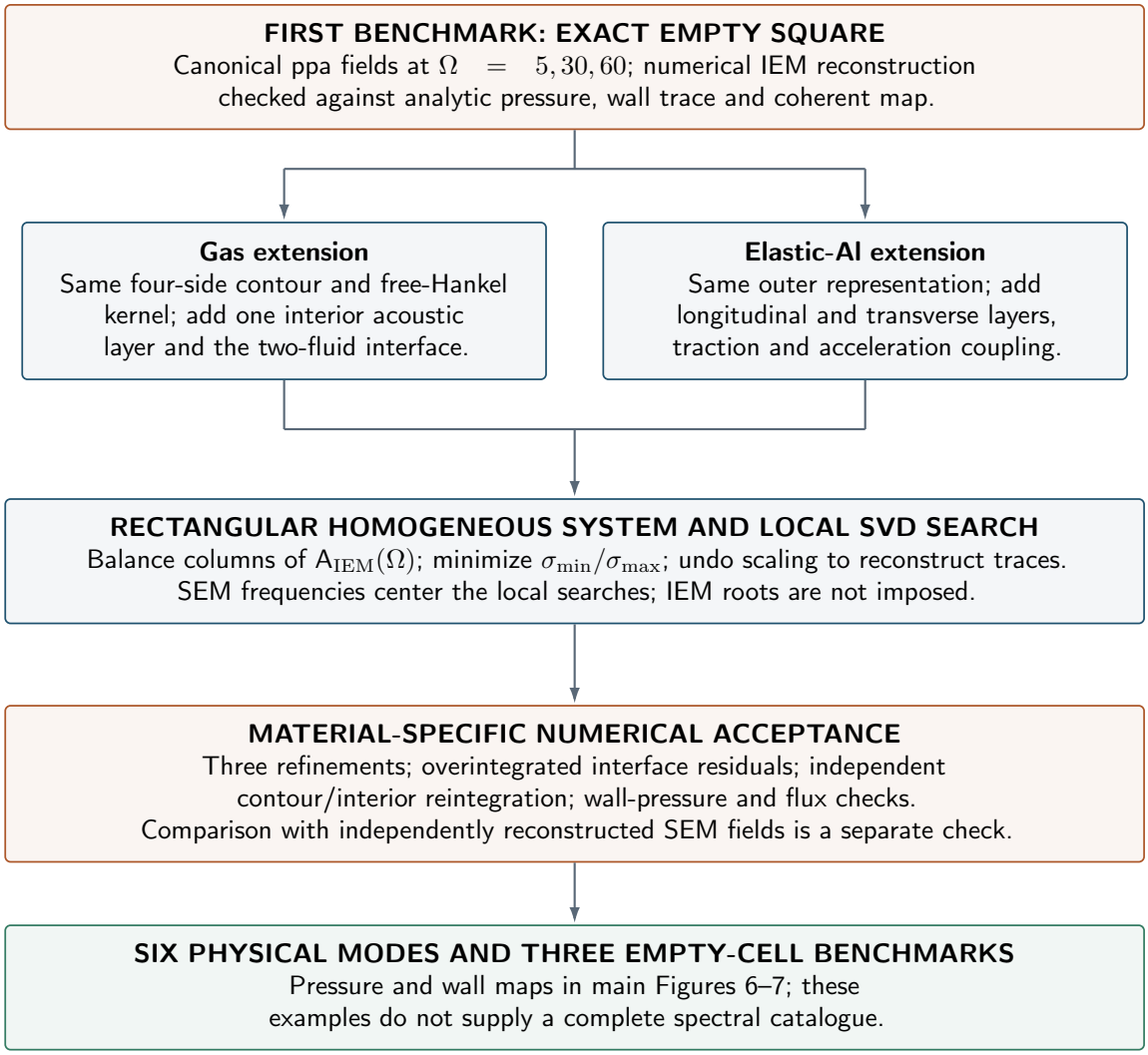


Diagram D4. IEM formulation, selected-root extraction and independent checks. The same closed-contour Green identity underlies empty, gas and elastic cases; only the latter two have circular material coupling. SVD treats the full rectangular system and supplies the coefficients after column scaling is undone. The final output is an atlas of selected modes, not an interval-wide IEM root enumeration.

Both atlases use the twelve panel identities in `shared/representative_modes.json`: analytic square, inclusion-free IEM, gas IEM and elastic-aluminum IEM, in that order. The

six physical roots retain numerical precision; targets 5, 30 and 60 organize selection rather than replace those roots. The first two rows show the same three canonical square modes, so twelve panels do not mean twelve distinct physical eigenmodes. Membership in an inference window remains recorded in SM Table SXII.

B. Explicit analytic benchmark

The supplement gives the separation indices and degeneracy selection of the analytic row. Its canonical real pressure field can be written explicitly as

$$p_N(x,y) = \frac{\cos(Ny) - \cos(Nx)}{\sqrt{2}}, \qquad N \in \{5, 30, 60\}. \tag{1}$$

Here x, y are reduced Cartesian coordinates and N an integer benchmark frequency. This fixes the $(0, N)$ member of the exchange-antisymmetric even–even (ppa) square eigenspace. The other canonical pair is given in SM Sec. S.I.B. The amplitude is conventional and cancels in exterior squared-pressure normalization. Equation (1) describes the analytic control. The empty-IEM field is independently reconstructed from numerical traces and compared with this control; the gas and aluminum cases receive their material-interface checks in SM Sec. S.I.C.

C. Quarter-cell arrays and full images

In `F_06_pressure_modes/data.npz`, the `normalized_squared_pressure_quarter` key has shape $(12, 512, 512)$, with indices panel, y row and x column. Let S_β be the square array of panel $\beta = 0, \ldots, 11$. Its quarter-cell sampling is

$$x_i = \frac{\pi(i + 1/2)}{n}, \quad y_j = \frac{\pi(j + 1/2)}{n}, \quad i, j = 0, \ldots, n-1, \quad n = 512. \tag{2}$$

The indices i, j are spatial, not modal. These values give the normalized squared-pressure observable displayed in Fig. 6.

Let $\mathcal{R}_x$ reverse array columns and $\mathcal{R}_y$ reverse array rows. The image operation implemented by the exporter is

$$\mathsf{G}_\beta = \begin{pmatrix} \mathcal{R}_y\mathcal{R}_x\mathsf{S}_\beta & \mathcal{R}_y\mathsf{S}_\beta \\ \mathcal{R}_x\mathsf{S}_\beta & \mathsf{S}_\beta \end{pmatrix}. \tag{3}$$

G_β is the 1024 by 1024 full-cell array. The blocks follow array row order; the image origin is lower and the physical extent is $[-\pi, \pi]^2$. The script in `F_06_pressure_modes` reflects each quarter array, applies the common color scale and material masks, arranges the twelve panels and exports the atlas. Material panels receive a circular cutout and outline. Empty-cell panels retain the complete saved field; the metadata radius identifies the common normalization region. The exporter preserves publication raster sampling, panel permutation and the saved numerical values.

D. Wall-map coordinates

The `H` array in `F_07_wall_husimi/data.npz` has shape $(12, 512, 512)$, indexed by panel, η row and q column. It contains the evaluated normalized wall density. The script in `F_07_wall_husimi` multiplies each map by 4π, arranges the same twelve panel identities and displays the common scale from 0 to 8, with saturation above 8. The saved coordinate vectors use

$$\begin{aligned} q_i &= \frac{2\pi i}{n_q}, \qquad \eta_j = -1 + \frac{2(j+1/2)}{n_\eta}, \\ n_q &= n_\eta = 512, \qquad i, j = 0, \ldots, 511. \end{aligned} \tag{4}$$

q is reduced periodic wall position and η normalized tangential wavenumber. The q samples start at zero and exclude the repeated endpoint; the η samples are strip midpoints. Reuse should read these vectors, not infer them from image pixel centers. The identically named ray-section coordinate in Fig. S1 has a different physical definition. The field, panel order and wall-density normalization follow article Sec. VI.B and its Figs. 6–7.

V. COMPLETE-ENSEMBLE SPATIAL RECORDS

The complete-ensemble records connect spatial observables with the same declared sectors and frequency windows used in the article. The physical estimators are in article Sec. VI and SM Sec. S.V.

Figure 8 is generated by `F_08_central_acf_distances`. The script groups every modal joint-direction distance by system and sector, checks the stored means and counts, and plots all modal points with the means and saved descriptive 95% whole-mode bootstrap intervals.

The central ensembles contain 1469 aluminum and 1508 square modes in $26 < \Omega \leq 34$ (article Sec. VI.C).

Figure S4 is generated by `S_04_central_acf_curves`. For those same central ensembles, the script displays the saved mean pressure and connected squared-pressure autocorrelation functions (ACFs), pointwise descriptive bootstrap bands and prescribed random-wave reference curves. Directions are averaged for this display, while the Fig. 8 distances retain both directions (SM Sec. S.V.B).

Figures S5 and S6 are generated by `S_05_low_frequency_acf_curves` and `S_06_low_frequency_acf_distances`, respectively. The first script plots the mean curves, pointwise bands and Bessel references; the second groups per-mode joint-direction distances and displays their means and saved bootstrap intervals. Both use the matched $10 < \Omega \leq 18$ ensembles: 688 gas, 691 aluminum and 704 canonical square modes, kept separate by sector (SM Sec. S.V.C).

Figure S7 is generated by `S_07_high_frequency_acf`. Its script combines mean ACF curves and their pointwise bands in the upper panels with every modal distance, means and saved descriptive intervals in the lower panels. It checks agreement between the mode counts, tabulated means and modal values. The common window $60 < \Omega \leq 61$, ppa, contains 48 gas, 47 aluminum and 49 square modes (SM Sec. S.V.D).

Figure 9 is generated by `F_09_directional_mixing`. The script groups the complete central wall records by system and sector and computes their empirical 10th, 25th, 50th, 75th and 90th percentiles. It plots medians, quartiles and 10–90% between-mode ranges for coherent participation and effective standing-wave channels (article Sec. VI.D and SM Sec. S.V.E).

a. Curve arrays. The ACF archives retain 161 normalized lags in `s`, the spatial separation measured in local air wavelengths. For each system/sector, a `_mean`, `_low` or `_high` array has shape $(2, 161)$. Index 0 selects pressure and index 1 connected squared pressure. Directions have already been combined for these display curves. The `pressure_reference` and `intensity_reference` vectors are stored comparison curves.

b. Modal identities and observables. A `modes.csv` row identifies its system, symmetry sector and reduced frequency. Where present, `mode_rank_in_window` is a window-specific

rank and `source_column_zero_based` is a zero-based upstream array index. Central distance and wall tables describe the same ensembles; joins use system, sector and modal frequency/identity together.

The `omega` column stores the reduced frequency Ω. The label `connected_intensity` denotes connected squared pressure. In the wall table, `participation` denotes coherent-map participation P_H and `channel_participation` the standing-wave channel count N_{ch}. The article defines the conversion to physical units for a chosen dimensional period.

c. Intervals and retained resolution. Saved bootstrap limits in `summary.csv` describe uncertainty of ensemble means. Pointwise ACF bands, finite-reference spectral quantiles, between-mode quantiles and refinement envelopes retain their different meanings. The scripts summarize the saved modal values and display the associated interval records. Joint-direction distances retain directional information before display averaging. Curve archives and modal tables therefore describe complementary aspects of each ensemble.

VI. ENERGY AND INTERVENTION RECORDS

Figure 10 is generated by `F_10_rigid_interface_control`. Its script reads `data.npz` and plots two complementary controls. Panels (a), (b) display the saved gap-scaled frequency shifts, refinement envelopes and tangential-channel pairs for all 184 matched elastic/rigid modes in the central ppa window. Panel (c) displays the separate local planar-interface pressure-reflection phases versus incidence angle for gas and aluminum (article Sec. VI.E and SM Secs. S.VI.B–S.VI.C).

Figure S8 is generated by `S_08_elastic_energy`. Its `modes.csv` table contains modal energy forms, interface-work and balance checks, quadrature sensitivity and solid-energy fraction; `data.json` gives sector summaries. The script groups all 1469 central aluminum modes by sector, plots their stored solid-energy fractions against reduced frequency, and computes sector means, medians and 5–95% between-mode quantiles for the summary panel. The energy definitions and checks are in SM Sec. S.VI.A.

Diagram D5 retains the logical role of these controls. It separates the ray theorem, finite-window wave evidence and mechanism constraints, without assigning a dynamical classification to every mode or treating reference-model rejection as a standalone test for or against chaos.

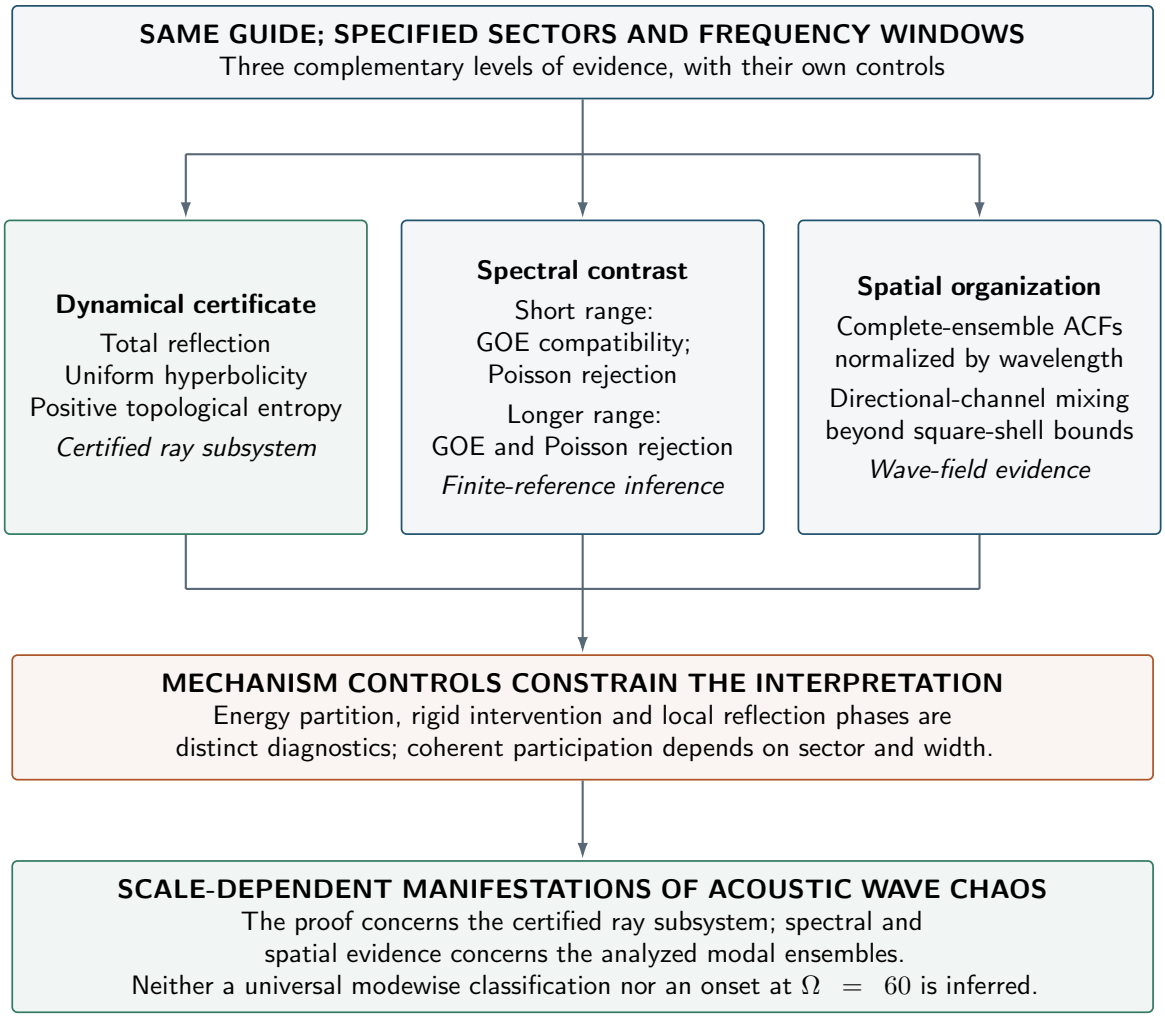


Diagram D5. Evidence and interpretation for acoustic wave chaos. The short-/long-range spectral contrast is resolved for the central aluminum and expanded high-frequency physical catalogues; the narrow band retains its separate power-limited results. Spatial diagnostics use complete modal ensembles and an integrable control. Energy and boundary interventions inform the mechanism without serving as separate chaos classifiers. The diagram makes the scopes explicit rather than asserting global chaos or an onset frequency.

VII. FIGURE-TO-FILE CORRESPONDENCE AND REUSABLE UNIT

Tables I and II list the sixteen directories in publication order. Each contains `plot.py`. The root `main.py` reads `FIGURE_INDEX.json` and dispatches plotting scripts; `plot_support.py` supplies common readers, styling and PDF export. The reusable unit is the complete `PnCW` root from the public PnCW repository, including its figure directories and `shared/` records. CSV files retain named columns, NPZ files contain compressed numerical arrays, and JSON

files retain metadata or accepted summaries. Figures 4 and S3 both read `shared/central_spectrum.json`. The two atlases share `shared/representative_modes.json`.

TABLE I. Result-directory and data correspondence: main figures. Each named directory contains `plot.py` and groups inputs by published observable.

Figure	Directory	Inputs and plotted quantities
3	`F_03_ray_certificate`	`data.json`: certified angles, sampled instability ranges and the symbolic adjacency matrix.
4	`F_04_spectral_scale_contrast`	Shared `central_spectrum.json`: adjacent-gap mean and number variance at $L_s = 8$, with finite-reference quantiles.
5	`F_05_high_frequency_spectra`	`curves.csv`, `data.json`: ratio CDFs and both-unfolding number variance in $(58, 62]$.
6	`F_06_pressure_modes`	`data.npz` and shared `representative_modes.json`: twelve squared-pressure panels, with analytic and inclusion-free IEM benchmarks first.
7	`F_07_wall_husimi`	`data.npz` and the same shared mode metadata: twelve normalized wall maps for the Figure 6 modes.
8	`F_08_central_acf_distances`	`modes.csv`, `summary.csv`: every central-mode joint-direction distance and saved descriptive mean intervals.
9	`F_09_directional_mixing`	`modes.csv`: complete central P_H and N_{ch} values; medians, quartiles and 10–90% ranges.
10	`F_10_rigid_interface_control`	`data.npz`: 184 elastic/rigid pairs, refinement envelopes, channel counts and planar reflection-phase samples.

TABLE II. Result-directory and data correspondence: supplementary figures. Each named directory contains `plot.py` and groups inputs by published observable.

Figure	Directory	Inputs and plotted quantities
S1	`S_01_periodic_ray_section`	`collisions.csv`, `data.json`: 336 collision records, the stored triangle itinerary and total-reflection thresholds.
S2	`S_02_numerical_validation`	`order.csv`, `mps.csv`, `acf.csv`: order checks, independent MPS samples and the complete high-frequency ACF-grid audit.
S3	`S_03_number_variance`	Shared `central_spectrum.json`: five sectors, four lengths and two unfoldings, with finite-reference bands.
S4	`S_04_central_acf_curves`	`data.npz`, `data.json`: central mean ACFs, saved pointwise bootstrap bands and prescribed Bessel references.
S5	`S_05_low_frequency_acf_curves`	`data.npz`, `data.json`: lower-window ACFs for gas, aluminum and the square.
S6	`S_06_low_frequency_acf_distances`	`modes.csv`, `summary.csv`: lower-window per-mode distances and saved mean intervals.
S7	`S_07_high_frequency_acf`	`data.npz`, `modes.csv`, `summary.csv`: high-window ACFs and all distances for 48 gas, 47 aluminum and 49 square modes.
S8	`S_08_elastic_energy`	`modes.csv`, `data.json`: all 1469 central solid-energy fractions and sector summaries.

VIII. TERMINAL EXECUTION AND PDF EXPORT

Download the package from the public PnCW repository using `Code > Download ZIP`, extract the archive and open a terminal in the extracted package directory. From the `PnCW` root on macOS, run the local setup script. It locates Python 3.12, recreates the machine-specific `.venv` environment and installs the NumPy/Matplotlib versions pinned in `requirements.txt`:

```
sh setup_macos.sh
```

If Python 3.12 is missing, install it once with Homebrew using `brew install python@3.12`, then rerun the script. The same script accepts `PNCW_PYTHON` to specify an interpreter outside the shell path. A virtual environment copied from another computer must be recreated locally. Windows installation commands are in `README.txt`. The plotting scripts read their local data files and use the shared readers, styles and export settings.

List the mapping and render all sixteen PDFs sequentially:

```
.venv/bin/python main.py --list
.venv/bin/python main.py --output generated
```

For selected figures or one local script:

```
.venv/bin/python main.py --figure 06 --figure 07 \
  --output selected
.venv/bin/python F_06_pressure_modes/plot.py \
  --output selected
```

Outputs are `Figure_03.pdf`–`Figure_10.pdf`, `Figure_S01.pdf`–`Figure_S08.pdf` and `PLOT_RUN.json`. The runner executes the selected plotting scripts sequentially and sets numerical-library threads to one. Same-name PDFs are replaced and numerical inputs are preserved. On Windows use `.venv/Scripts/python.exe`.

For each selected directory, `main.py` launches `plot.py`, checks that its PDF has been created and records the figure identifier, output filename and Python version in `PLOT_RUN.json`. Each individual script imports the shared helper and reads its applicable local or shared data records.